

# Unified Tool Surface and Extended Bag Relational Algebra

One immutable artifact system across Direct, Atomic, and Hybrid modes

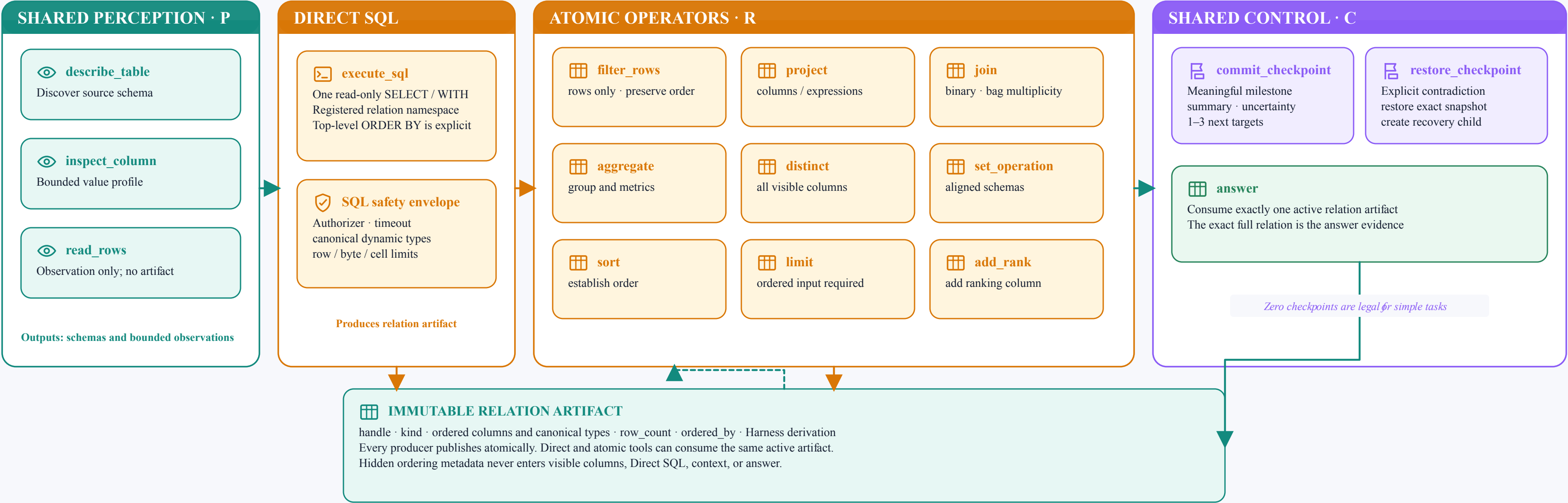
DIAGNOSTIC ONLY

DIRECT = P + SQL + C

ATOMIC = P + R + C

HYBRID = DIRECT ∪ ATOMIC

Current experiment mode · SQL and atomic artifacts interoperate



F04 · Mode surfaces

$$\mathcal{I}_D = \mathcal{P} \cup \{\text{execute\_sql}\} \cup \mathcal{C}, \quad \mathcal{I}_A = \mathcal{P} \cup \mathcal{R} \cup \mathcal{C}, \quad \mathcal{I}_H = \mathcal{I}_D \cup \mathcal{I}_A$$

F05 · Artifact contract

$$A = \langle h, k, \langle c_i : \tau_i \rangle_{i=1}^m, n, \omega, d \rangle$$

F06 · Extended relational algebra

$$\sigma_p(R), \quad \pi_{\langle e_i \rightarrow a_i \rangle_{i=1}^m}(R), \quad R \bowtie_{\theta} S, \quad \gamma_{G; \langle f_i(c_i) \rightarrow a_i \rangle}(R), \quad \tau_K(R)$$

F07 · Bag join multiplicity

$$\mu_{R \bowtie_{\theta} S}(r, s) = \mu_R(r) \mu_S(s) \mathbf{1}[\theta(r, s)]$$